

- Equation of Straight line

$$y = mx + c$$

$$ax + by + c = 0$$

$$W^T x + c = 0$$

- Use of W.V (to know if the data point is on +ve or -ve side).
- Sigmoid Activation Function (to shrink the data from 0 to 1)
- Entropy Gini Impurity

Components required to create straight line

- x and y point
- Slope
- Intercept

Components required to create straight line

- $x = 1$
- $y = 1$
- Slope = 1.5

$$y = mx + c$$

## Slope

$$m = \frac{\text{Change in Y}}{\text{Change in X}}$$

$$\text{Slope } (m) = \frac{Y_2 - Y_1}{X_2 - X_1}$$

X      Y

2      10

4      20

$$Y_1 = 10$$

$$Y_2 = 20$$

$$X_1 = 2$$

$$X_2 = 4$$

$$m = \frac{20 - 10}{4 - 2}$$

$$m = \frac{10}{2}$$

$$m = 5$$

# Equation of Straight Line

$$y = m x + c$$

**m** (Slope or Coefficient) = slope of the line

**c** Intercept (where the best-fit line meets the Y axis).

$$M=1.3$$

$$C=0$$

| X   | m x + c = y          |
|-----|----------------------|
| 1.0 | (1.3 * 1) + 0 = 1.3  |
| 3.0 | (1.3 * 3) + 0 = 3.9  |
| 4.0 | (1.3 * 4) + 0 = 5.2  |
| 6.0 | (1.3 * 6) + 0 = 7.8  |
| 7.0 | (1.3 * 7) + 0 = 9.1  |
| 8.0 | (1.3 * 8) + 0 = 10.4 |

$$M=1.2$$

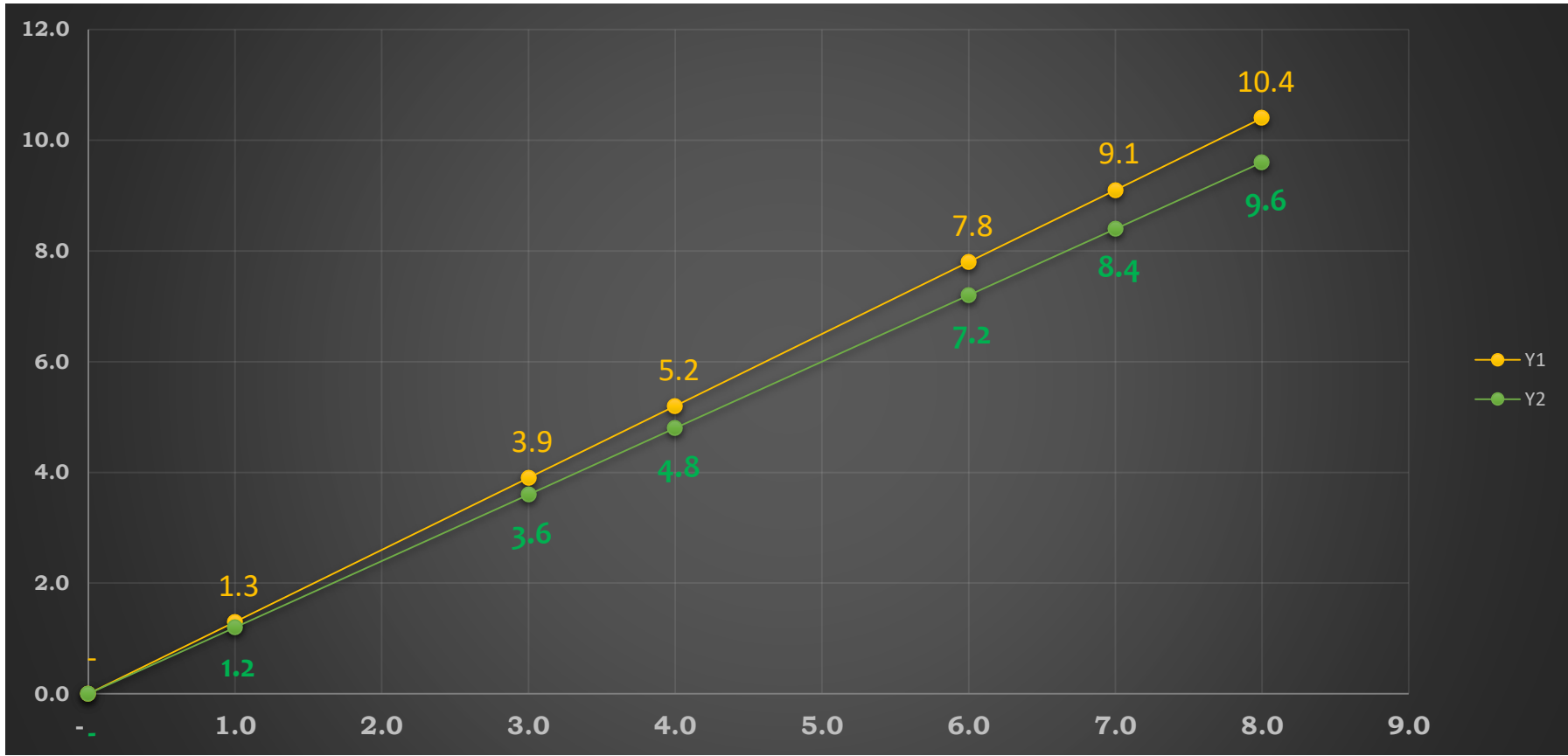
$$C=0$$

| m x + c = y         |
|---------------------|
| (1.2 * 1) + 0 = 1.2 |
| (1.2 * 3) + 0 = 3.6 |
| (1.2 * 4) + 0 = 4.8 |
| (1.2 * 6) + 0 = 7.2 |
| (1.2 * 7) + 0 = 8.4 |
| (1.2 * 8) + 0 = 9.6 |

M=1.3  
C=0

| x   | y_1  | Y_2  |
|-----|------|------|
| 1.0 | 1.3  | 01.2 |
| 3.0 | 3.9  | 3.6  |
| 4.0 | 5.2  | 4.8  |
| 6.0 | 7.8  | 7.2  |
| 7.0 | 9.1  | 8.4  |
| 8.0 | 10.4 | 9.6  |

M=1.2  
C=0



## *Equation of Straight Line*

$$y = \mathbf{m} \mathbf{x} + \mathbf{c}$$

$$y = \mathbf{m} \mathbf{x} + \mathbf{c}$$

## *Equation of Straight Line*

$$ax + by + c = 0$$

$$by + c = -ax$$

$$by = -ax + (-c)$$

$$y = \frac{-\mathbf{a}}{\mathbf{b}} \mathbf{x} + \frac{-\mathbf{c}}{\mathbf{b}}$$

$\mathbf{m}$  OR  $\frac{-\mathbf{a}}{\mathbf{b}}$  = (Slope or Coefficient) slope of the line

$\mathbf{c}$  OR  $\frac{-\mathbf{c}}{\mathbf{b}}$  = Intercept (where the best-fit line meets the Y axis).

$$-1x + 2y - 2 = 0$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 2 = -1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y - 1 = 0$$

$$y = m x + c$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y = 1 + 2$$

$$-1x + 2y = +3$$

$$-1x + 2y = 3$$

$$2y = 1x + 3$$

$$2y = 1x + 3$$

$$y = \frac{1}{2} x + \frac{3}{2}$$

$$y = \frac{1}{2} x + \frac{3}{2}$$

$$y = 0.5 x + 1.5$$

$$y = 0.5 x + 1.5$$

$$-1x + 2y - 2 = -1$$

$$-1x + 2y = -1 + 2$$

$$-1x + 2y = 1$$

$$2y = 1x + 1$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

$$y = 0.5x + 0.5$$

$$-1x + 2y - 1 = 0$$

$$-1x + 2y = 1$$

$$2y = 1x + 1$$

$$2y = 1x + 1$$

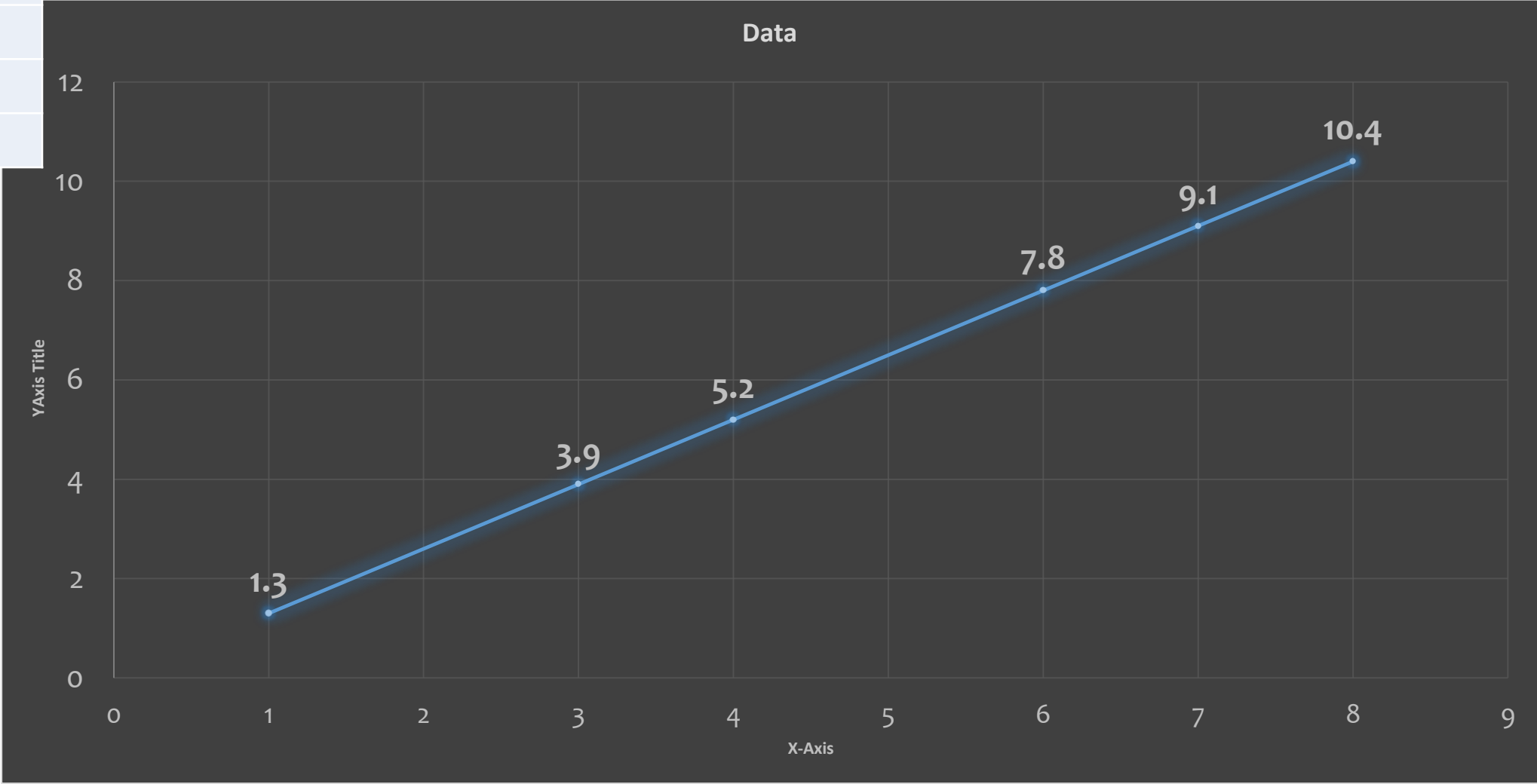
$$y = \frac{1}{2}x + \frac{1}{2}$$

$$y = 0.5x + 0.5$$





| X     | Y        |
|-------|----------|
| X = 1 | y = 1.3  |
| X = 3 | y = 3.9  |
| X = 4 | y = 5.2  |
| X = 6 | y = 7.8  |
| X = 7 | y = 9.1  |
| X = 8 | y = 10.4 |



## Equation of Straight Line

$$ax + by + c = 0$$

$$y = m x + c$$

$$X=[1,3,4,6,7,8]$$

If

$$a = 1.3$$

$$b = -1$$

$$c = 0$$

## Equation of Straight Line

$$ax + by + c = 0$$

$$1.3(1) + (-1)y + 0 = 0$$

$$1.3 - y + 0 = 0$$

$$1.3 - y = 0$$

$$-y = 0 - 1.3$$

$$-y = -1.3$$

$$y = 1.3$$

| x | (ax+by+c=0) (Substitute values) | Output       |
|---|---------------------------------|--------------|
| 1 | $1.3(1) - y + 0 = 0$            | $(y = 1.3)$  |
| 3 | $1.3(3) - y + 0 = 0$            | $(y = 3.9)$  |
| 4 | $1.3(4) - y + 0 = 0$            | $(y = 5.2)$  |
| 6 | $1.3(6) - y + 0 = 0$            | $(y = 7.8)$  |
| 7 | $1.3(7) - y + 0 = 0$            | $(y = 9.1)$  |
| 8 | $1.3(8) - y + 0 = 0$            | $(y = 10.4)$ |